This report outlines the development of a **Multimodal Regression Pipeline** designed to predict property market values by integrating traditional tabular data with high-dimensional embeddings extracted from satellite imagery

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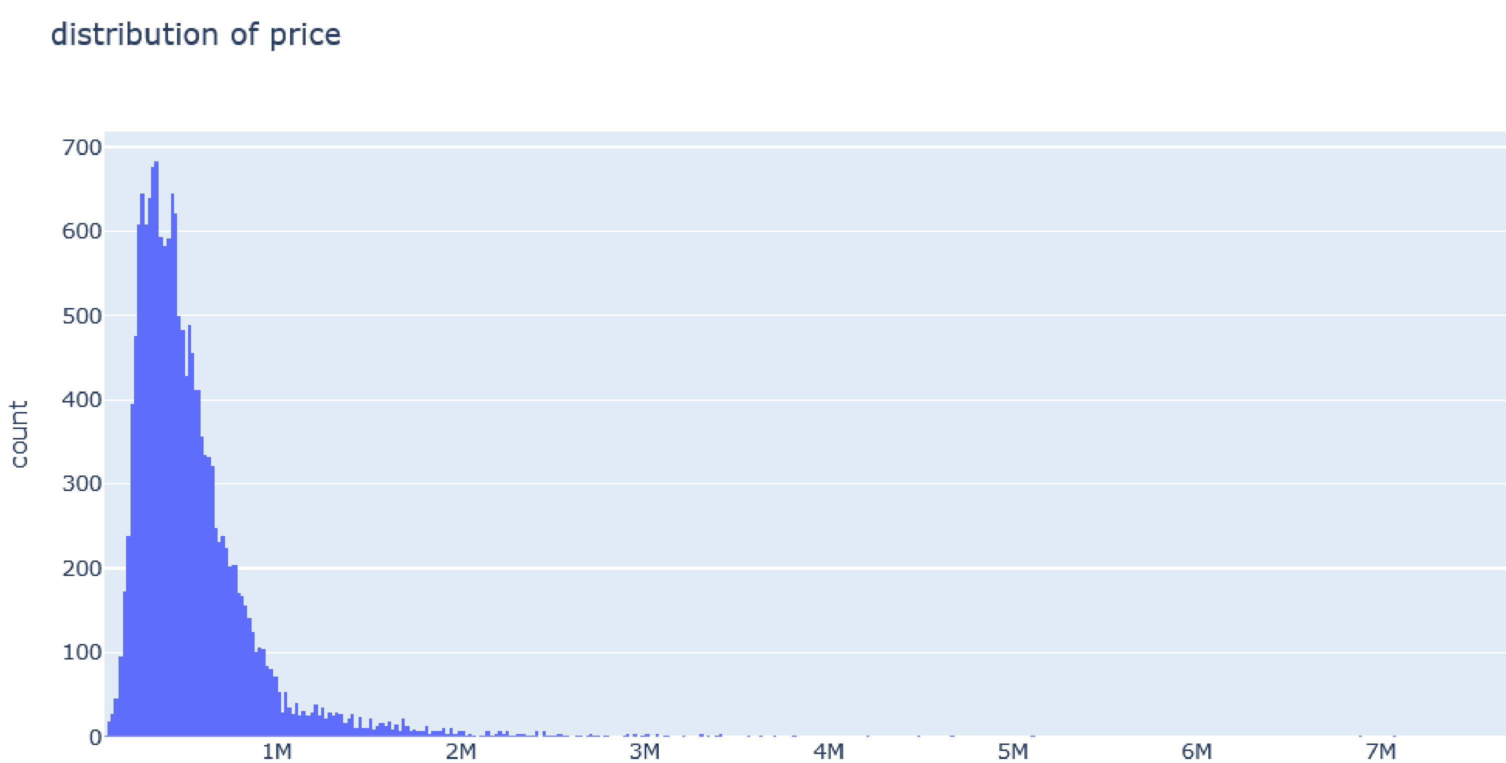
Executive Summary

The primary objective of this project was to move beyond standard house-price modeling by capturing the "environmental context" of a property. By programmatically acquiring satellite images based on latitude and longitude, we successfully integrated neighborhood characteristics—such as green cover and road density—into a unified XGBoost framework.

Our approach utilized a **late-fusion architecture** to combine structured and unstructured data

- **Tabular Data**: Historical housing data including features like sqft_living, grade, and condition etc
- **Visual Data**: Satellite images were fetched programmatically using coordinates.

Exploratory Data Analysis (EDA)



For price we can use log transformation, although it does not mater much in case of XGBoost (i.e tree best methods).

Geospatial Insights

Analysis revealed that visual factors significantly influence valuation:

- **Waterfront Proximity**: Properties with a waterfront score of 1 showed a distinct price premium.
- **Neighborhood Density**: Features like sqft_living15 confirmed that a property's value is heavily dictated by the scale of its immediate neighbors.

Feature Engineering & Signal Extraction

In real estate analytics, the raw data rarely captures the full economic story. We implemented a sophisticated engineering pipeline to transform raw timestamps, locations, and images into high-signal predictors.

1. Time Handling

Time is a critical factor in finance and economics. We decomposed the raw date string into cyclical and linear numeric components:

- **Sale Year & Month**: Extracted to capture market inflation and seasonal trends in the housing market.
- **Day of Week/Month**: Engineered to identify potential patterns in closing dates or listing behaviors.
- **Significance**: This transformation allows the XGBoost model to treat time as a multi-dimensional feature rather than a simple categorical string.

2. Zipcode Target Encoding

Location is the primary driver of property value. Instead of treating zipcode as a high-cardinality label, we used **Target Encoding**:

- **Mechanism**: We mapped each zipcode to its median property price derived from the training set.
- **Benefit**: This creates a continuous "Location Index" that quantifies the relative wealth and desirability of an area, effectively condensing complex geospatial economics into a single numeric feature.

3. House Age & Effective Built Year

The age of a structure is not always the year it was built. We calculated **Effective House Age**:

- **Logic**: If a property was renovated (yr_renovated > 0), the "effective year" becomes the renovation year; otherwise, it remains the original build year.
- **Significance**: This accurately reflects the modern condition and "wear-and-tear" of the asset, which is a major factor in construction quality grading.

4. Transfer Learning & Visual Embedding Extraction

We leveraged **Transfer Learning** to interpret the satellite imagery without requiring a massive labeled image dataset:

- **Pre-trained Backbone**: We utilized a CNN (EfficientNet-B4) pre-trained on millions of natural images to act as a "feature extractor".
- **Mechanism**: By removing the final classification layer, we extracted the "global average pooling" layer as a high-dimensional vector (embedding) for each property.
- **Logic**: This allows our model to inherit "visual common sense"—understanding shapes, textures, and borders—and apply that knowledge to recognize property characteristics like "curb appeal"

5. Visual Signal Compression: PCA on Image Encodings

Our pipeline extracted 1,792 features from each satellite image using a pre-trained CNN. To prevent the model from over-fitting on noise (the "Curse of Dimensionality"), we applied **Principal Component Analysis (PCA)**:

- **Variance Retention**: We targeted 40% of the total variance, which compressed the 1,792 features down to the 12 most significant "principal components."
- **Outcome**: This reduced the feature space by over 99% while retaining the core visual markers of the neighborhood, such as road density and proximity to water.

Results on training data with CV

We compared the performance of two distinct modeling approaches:

Model Version	Features Used	RMSE	R^2 Score
Baseline	Tabular Only	*107575.09*	*0.895*
Multimodal	Tabular + PCA Image Embeddings	*111736.39*	*0.885*

Conclusion

By fusing satellite imagery with traditional real estate metrics, this pipeline provides a more holistic valuation framework. Future work will involve fine-tuning the CNN backbone specifically on satellite-view datasets to improve the granularity of visual feature extraction